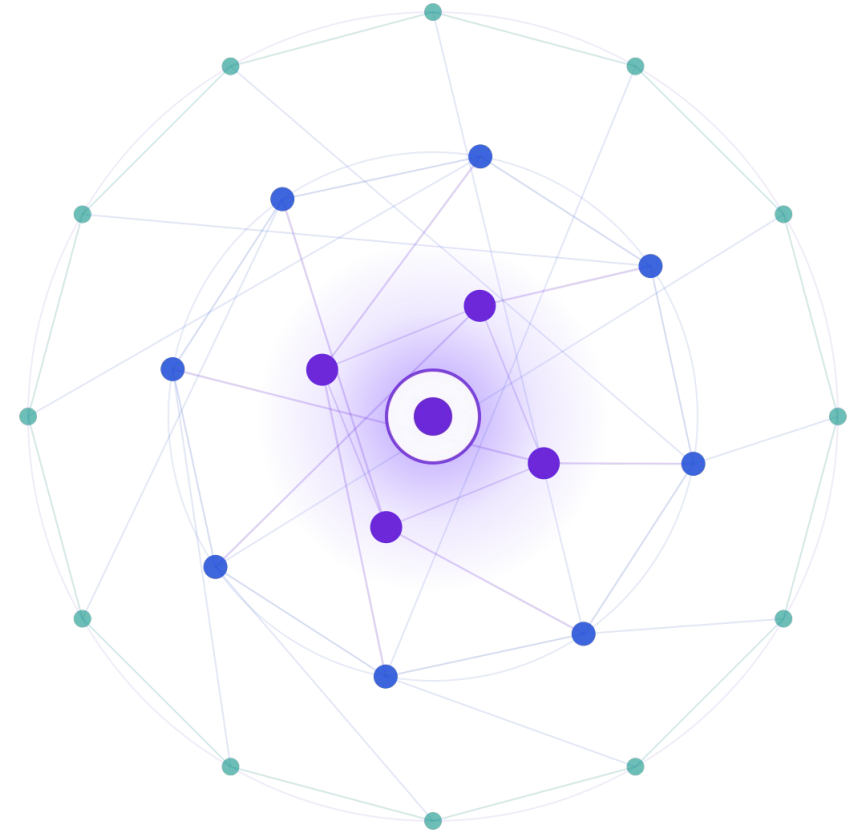


Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID – 1 (Agentic Code Intelligence)
 - Team Name – Trace
 - College Name – Thapar Institute of Engineering and Technology, Patiala
 - Member – Abhavya Jha – ajha2_be24@thapar.edu
-
- Submission GitHub link – <https://github.com/Ajha2005/samsung-prism>



Theme

Theme 1 — Agentic Code Intelligence

Given a natural-language query and a library of code snippets, return the snippets ranked by relevance — pure retrieval, no generation or explanation.

Scored by MTEB's NDCG@10 and MRR on the ColR AppsRetrieval test split: a real, objective leaderboard number decides the outcome — not a live demo or a jury.

THE SCOREBOARD · SUBMITTED: mpnet-512

NDCG@10 = 0.0861

MRR@10 = 0.0733 · +30.1% over baseline

THE DISCIPLINE THAT PRODUCED IT

9

configs measured

7

dropped

1

kept: mpnet-512

77

offline tests

1

CPU-only, minimal GPU

Innovation has to come from representation, not scale

2

No LLM in the ranking loop

Retrieval must beat generation on speed on a thousand-snippet corpus

3

Cross-version retrieval (P1)

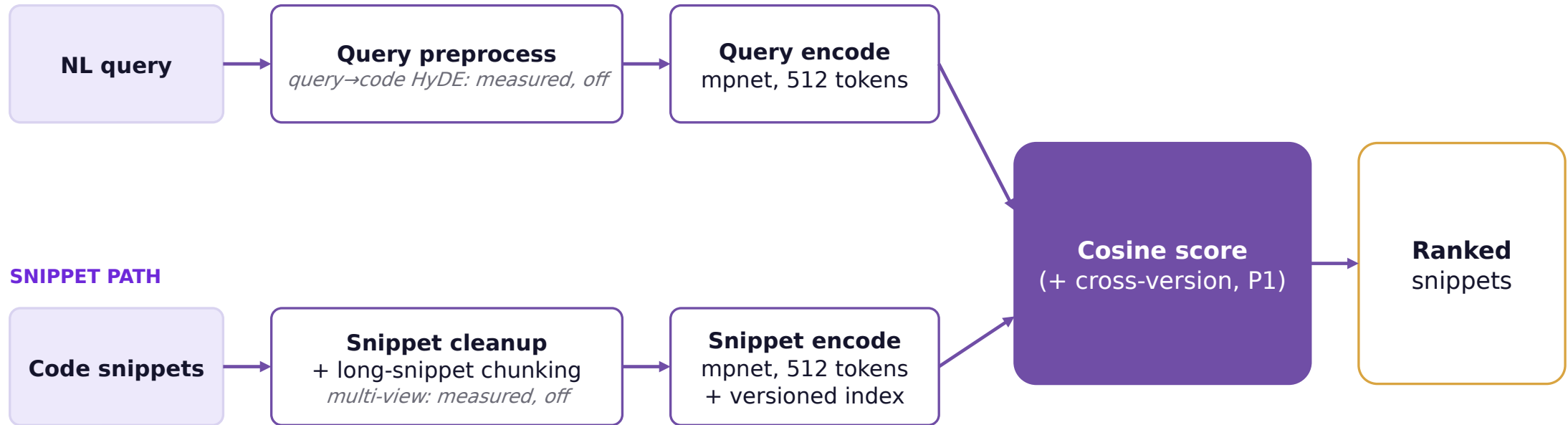
The index must rebuild cheaply as code changes, by design from day one

Existing Solutions & Gaps

WHAT TEAMS USUALLY TRY		WHERE IT BREAKS DOWN		HOW WE DESIGNED FOR THIS
1 Dense embedding + cosine similarity with an off-the-shelf text model	>	NL and code sit in different “languages” — behavioral queries blur	>	Code-flavored HyDE: compile query → deterministic code sketch, then embed code-vs-code
2 LLM re-ranking on top of a first-pass retriever	>	Corpus in the thousands → per-query LLM ranking slower than retrieval itself	>	No LLM in the ranking loop; any LLM use is offline pre-processing, cached
3 One raw-text view per snippet, straight into the encoder	>	Raw text blurs the structure and identifiers usage & structural queries need	>	Multi-view snippet embedding: raw + AST-normalized + identifier bag, fused
4 A single index built once for one code version	>	Near-identical versions become indistinguishable; every change rebuilds all	>	Content-addressed versioned index + stable-core / version-delta representation

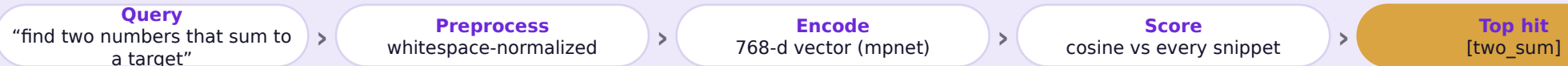
Our Solutions & Architecture Diagram

QUERY PATH



Submitted: all-mpnet-base-v2, 512-token window, CPU-only. Grey items are built and switchable by config, but lost on the real split and are off.

EXAMPLE TRACE — one real query through the pipeline



Demo & Product Walkthrough



```
$ python -m prism.cli demo --backend hashing  
[prism-demo] embedding backend: offline HashingBackend
```

Q: find two numbers that sum to a target

1. [two_sum] score=0.3810
2. [binary_search] score=0.1483

mean query latency: 0.03 ms (36,000+ q/s)

0.03 ms

mean query latency (offline demo)

36,000+

queries / second (offline demo)

3 more behavioural queries, top-1 hits on the same corpus

Q “check if a string is a palindrome”

► **is_palindrome**

score 0.44

Q “shortest path in an unweighted graph”

► **bfs**

score 0.15

Q “count word frequencies in a text”

► **word_count**

score 0.48

Offline demo (hashing backend, no downloads): same pipeline and ranking code as the leaderboard run — the submitted model swaps in by config.

Tools and tech stack used

1

Core ML & evaluation

MTEB is the same library the leaderboard runs; picking it removes any doubt about metric parity.

Python 3.11 · PyTorch (CPU) · sentence-transformers · MTEB (official scorer)

2

Retrieval signals

Stdlib `ast` avoids a third-party parser dependency — the eval box does not need to install one.

NumPy · rank-bm25 (sparse/BM25) · Python `ast` (structural normalization)

3

Reproducibility & infra

The default `docker run` runs the whole test suite and demo with no HuggingFace round-trip needed.

Docker · pytest — 77 offline tests · GitHub Actions-ready CLI

BY THE NUMBERS

77

offline tests

9

configs measured

1

docker run

~0.04 ms

offline demo latency

100%

CPU inference

Impact & Use case

1

Faster onboarding

For: new engineers

Search unfamiliar code by describing behavior, not memorized function names.

EXAMPLE QUERY

"where do we retry a failed payment?"

► *Days-to-productivity → hours.*

2

Agentic coding backend

For: AI coding agents

A fast retrieval step an agent calls before it generates or edits code.

RETRIEVAL BUDGET

one encoder pass per query, no LLM in the loop.

► *Retrieval faster than generation, by design.*

3

Code review & audit

For: reviewers / security

Locate every place a pattern or behavior appears — not just a grep match.

EXAMPLE QUERY

"all endpoints that write user email unmasked"

► *Finds behavioral clones grep misses.*

4

Legacy modernization

For: platform teams (P1)

Search & compare behavior across many historical versions of the same code.

EXAMPLE QUERY

"auth check pre-v3.2 vs post-v3.2"

► *Cross-version rebuilds cheap by design.*

Innovation highlights, results and limitations

Innovation highlights

Built and rigorously measured — kept only where the number agreed (see results →)

- Query→code compilation (code-flavored HyDE), LLM-free at eval time
- Multi-view snippet embedding: raw + AST-normalized + identifier bag
- Keyword front-loading: salient signals survive a 256-token tokenizer cap
- Stable-core + version-delta representation for cross-version retrieval
- Content-addressed versioned index — rebuilds only what changed

THE DISCIPLINE THAT PRODUCED THAT NUMBER

Propose
one config

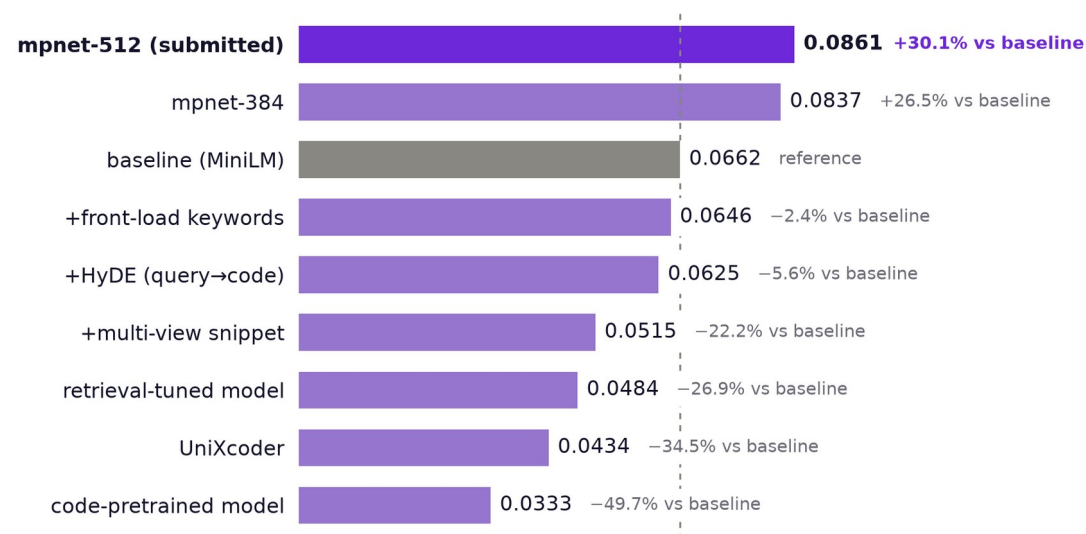
Measure
NDCG@10 on
real split

Compare
vs live baseline

Keep or drop
no drift, ever

NDCG@10, real CoIR AppsRetrieval split · 9 configs, 1 kept

Submitted: mpnet-512 — MRR@10 0.0733 (baseline 0.0561).



Limitations — stated up front

- CPU-only caps model size by design — the trade this theme asks for
- MiniLM→mpnet mixes size and window; 384→512 alone added +2.9%
- HyDE, multi-view, front-load not yet re-measured on top of mpnet

What's next

WHAT THE ABLATION TAUGHT US

Nine configs on the real split. Only the general-purpose recipe at 5× the size beat baseline (mpnet, +26.5%), and a 512-token window added +2.9% more. Every specialized swap (retrieval-tuned, CodeSearchNet, UniXcoder) lost — the roadmap follows that evidence.

1

Re-test on mpnet

HIGH

HyDE, multi-view and front-loading lost on MiniLM; a 5× larger encoder may use them differently.

2

Split size vs window

MED

384→512 alone added +2.9%; run mpnet at 256 tokens to isolate the gain from model size.

3

Validate at scale

MED

Prove cross-version retrieval on real multi-version repositories, not just the prototype set.

4

Publish the findings

LONG

Cross-version code retrieval is under-solved territory — PRISM cohorts have taken this to IEEE.

Every step above is measured the same way as everything before it — one change, one number, keep or revert.

Brownie points slide(differentiation)



Verified, not assumed

Proven end-to-end against a real mteb.evaluate() run.

EVIDENCE

tests/test_submission_format.py — 6 tests lock the leaderboard file shape byte-for-byte to the problem statement.



Discipline over guesswork

Every feature logged config → NDCG@10 → kept/dropped, every time.

EVIDENCE

docs/ablation_log.md — 9 real experiments, 7 dropped, 1 kept (+30.1%); every row reproducible from a config file.



Versioned from day one

Content-addressed index; cross-version retrieval never bolted on.

EVIDENCE

prism/index/versioned.py + stable-core / version-delta — the P1 requirement is a first-class feature.



Honest about limits

Limitations reported next to results — credibility over overselling.

EVIDENCE

docs/ablation_log.md — every failure named up front; a code-model swap losing 50% is documented, not hidden.

BY THE NUMBERS

9

real experiments

7

dropped, still in repo

77

offline tests

0

mocks in leaderboard eval

1

reproducible docker run

Checklist- Updated on Public GitHub

SUBMISSION CHECKLIST



Working prototype code

Public GitHub repo, tag PRISM_GENAI_HACKATHON_Y2026
github.com/Ajha2005/samsung-prism



README with reproducible setup

Docker one-command run + full CLI
`docker build -t prism-code-search . → docker run`



Demo video (≤ 5 min)

Recorded and linked in the submission form
[YouTube](#) / [Drive link](#) – see form



Presentation file (PPT + PDF)

This deck — 12 slides, speaker notes on every content slide
[Thapar_Trace_Submission_ppt.pptx](#) / [.pdf](#)



Leaderboard results JSON

Real CoIR AppsRetrieval, produced by `mteb.evaluate()`
[results/appsretrieval_results.json](#) (attached to release)



Ablation record

9 configs measured, 7 dropped, mpnet-512 kept (+30.1%)
[docs/ablation_log.md](#)

SUBMISSION AT A GLANCE

0.0861

NDCG@10 · CoIR AppsRetrieval · MTEB

VERIFIED

9

real ablations

+30.1%

over baseline

77

offline tests

100%

CPU inference

Tag · PRISM_GENAI_HACKATHON_Y2026

Thank you

Team Trace · Thapar Institute of Engineering and Technology

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github.com/Ajha2005/samsung-prism